Empirical Proof Record: REFUTED

2026-05-23T23:12:57.844884+00:00

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Proof ID: `efd4f11df9c451e0c3136f4d99e9619b…`
Created: 2026-05-23T23:12:57.844884+00:00

1. Claim

After the same documents in a different ORDER, the same identically-recognised query gets a different prime-address from Kimera (hysteresis) beyond its own run-noise, whereas a set-based retriever returns the identical ranking (order-blind by construction): order_hysteresis = mean(order_divergence) − mean(noise_floor) > 0.

- **Operationalisation:** Run 6 documents + 6 held-out probes through Kimera's entity target (batch) in order A, order A again (determinism control), and order B (shuffled). Per probe: order_divergence = 1 − Jaccard(prime_chain|A, prime_chain|B); noise_floor = 1 − Jaccard(prime_chain|A1, prime_chain|A2). order_hysteresis = mean(order_divergence) − mean(noise_floor). Baseline: TF-IDF retrieval order-divergence over the same docs/queries (= 0). VALIDATED iff order_hysteresis > 0 with a significant paired order_divergence > noise_floor test.
- **Threshold:** `order_hysteresis > 0.0 jaccard_distance`
- **H0:** H0: order_hysteresis <= 0 — Kimera's response does not depend on document order beyond run-noise; accumulation is commutative, like a set-based retriever
- **H1:** H1: order_hysteresis > 0 — the same query after the same documents in a different order gets a different prime-address; Kimera carries the ORDER of experience (hysteresis) where RAG carries only the inventory

2. Verdict

REFUTED — observed `0.0`
order_hysteresis +0.0000 = Kimera order-divergence 1.0000 − noise-floor 1.0000 over 6 probes (prime-address); RAG (TF-IDF) order-divergence 0.0000 (set-based, order-blind by construction); manifold-state order-effect coupling Δ=0.175958, mass Δ=6.289987 (noise coupling Δ=0.0)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
`O.comparison.order_hysteresis`	`order_hysteresis`	`0.0`	—	ophamin 0.115.2

4. Pre-registration

- Registered at: `2026-05-23T23:12:57.603543+00:00`
- Config hash: `e9e0949b499b8b4758262350a7ec8186…`
- Data hash: `3896f14d8eaa26af9e1b43a0bcd7ece7…`

5. Signature

`9c84b27aa3bf36c6e55999aa9dc403a101d4aa1c55cce38a7521fd401185bd86`